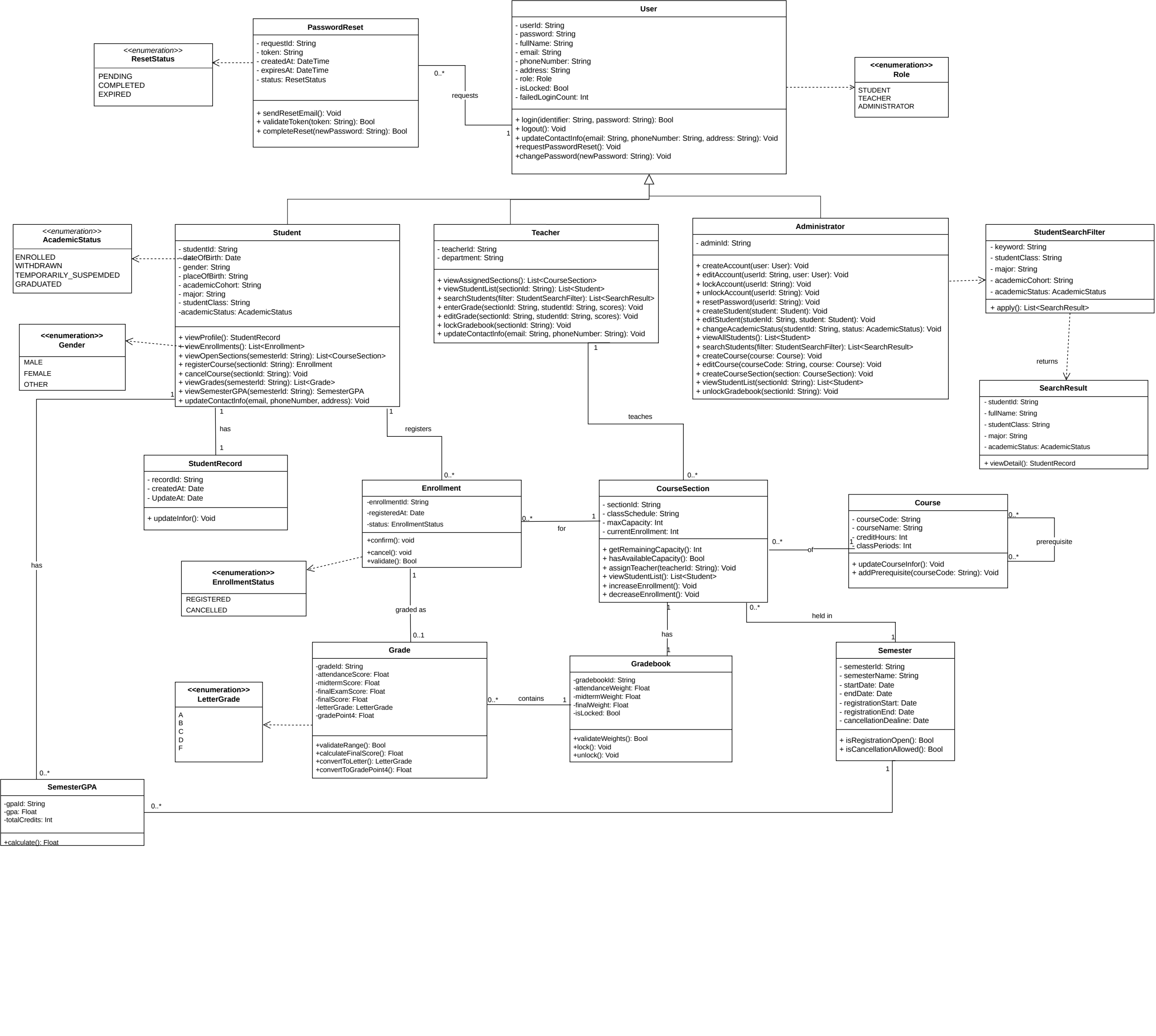

User
- userId: String
- password: String
- fullName: String
- email: String
- phoneNumber: String
- address: String
- role: Role
- isLocked: Bool
- failedLoginCount: Int
+ login(identifier: String, password: String): Bool
+ logout(): Void
+ updateContactInfo(email: String, phoneNumber, address: String): Void
+requestPasswordReset(): Void
+changePassword(newPassword: String): Void
PasswordReset
- requestId: String
- token: String
- createdAt: DateTime
- expiresAt: DateTime
- status: ResetStatus
+ sendResetEmail(): Void
+ validateToken(token: String): Bool
+ completeReset(newPassword: String): Bool
<<enumeration>>
ResetStatus
PENDING
COMPLETED
EXPIRED
0..*
requests
1
<<enumeration>>
Role
STUDENT
TEACHER
ADMINISTRATOR
Student
- studentId: String
- dateOfBirth: Date
- gender: String
- placeOfBirth: String
- academicCohort: String
- major: String
- studentClass: String
-academicStatus: AcademicStatus
+ viewProfile(): StudentRecord
+ viewEnrollments(): List<Enrollment>
+ viewOpenSections(semesterId: String): List<CourseSection>
+ registerCourse(sectionId: String): Enrollment
+ cancelCourse(sectionId: String): Void
+ viewGrades(semesterId: String): List<Grade>
+ viewSemesterGPA(semesterId: String): SemesterGPA
+ updateContactInfo(email, phoneNumber, address): Void
<<enumeration>>
AcademicStatus
ENROLLED
WITHDRAWN
TEMPORARILY_SUSPEMDED
GRADUATED
<<enumeration>>
Gender
MALE
FEMALE
OTHER
Teacher
- teacherId: String
- department: String
+ viewAssignedSections(): List<CourseSection>
+ viewStudentList(sectionId: String): List<Student>
+ searchStudents(filter: StudentSearchFilter): List<SearchResult>
+ enterGrade(sectionId: String, studentId: String, scores): Void
+ editGrade(sectionId: String, studentId: String, scores): Void
+ lockGradebook(sectionId: String): Void
+ updateContactInfo(email: String, phoneNumber: String): Void
Administrator
- adminId: String
+ createAccount(user: User): Void
+ editAccount(userId: String, user: User): Void
+ lockAccount(userId: String): Void
+ unlockAccount(userId: String): Void
+ resetPassword(userId: String): Void
+ createStudent(student: Student): Void
+ editStudent(studenId: String, student: Student): Void
+ changeAcademicStatus(studentId: String, status: AcademicStatus): Void
+ viewAllStudents(): List<Student>
+ searchStudents(filter: StudentSearchFilter): List<SearchResult>
+ createCourse(course: Course): Void
+ editCourse(courseCode: String, course: Course): Void
+ createCourseSection(section: CourseSection): Void
+ viewStudentList(sectionId: String): List<Student>
+ unlockGradebook(sectionId: String): Void
StudentSearchFilter
- keyword: String
- studentClass: String
- major: String
- academicCohort: String
- academicStatus: AcademicStatus
+ apply(): List<SearchResult>
returns
SearchResult
- studentId: String
- fullName: String
- studentClass: String
- major: String
- academicStatus: AcademicStatus
+ viewDetail(): StudentRecord
has
1
has
1
1
1
registers
1
teaches
1
0..*
0..*
StudentRecord
- recordId: String
- createdAt: Date
- UpdateAt: Date
+ updateInfor(): Void
Enrollment
-enrollmentId: String
-registeredAt: Date
-status: EnrollmentStatus
+confirm(): void
+cancel(): void
+validate(): Bool
0..*
for
1
CourseSection
- sectionId: String
- classSchedule: String
- maxCapacity: Int
- currentEnrollment: Int
+ getRemainingCapacity(): Int
+ hasAvailableCapacity(): Bool
+ assignTeacher(teacherId: String): Void
+ viewStudentList(): List<Student>
+ increaseEnrollment(): Void
+ decreaseEnrollment(): Void
0..*
of
1
Course
- courseCode: String
- courseName: String
- creditHours: Int
- classPeriods: Int
+ updateCourseInfor(): Void
+ addPrerequisite(courseCode: String): Void
0..*
prerequisite
0..*
<<enumeration>>
EnrollmentStatus
REGISTERED
CANCELLED
graded as
1
0..1
1
has
1
0..*
held in
1
Grade
-gradeId: String
-attendanceScore: Float
-midtermScore: Float
-finalExamScore: Float
-finalScore: Float
-letterGrade: LetterGrade
-gradePoint4: Float
+validateRange(): Bool
+calculateFinalScore(): Float
+convertToLetter(): LetterGrade
+convertToGradePoint4(): Float
<<enumeration>>
LetterGrade
A
B
C
D
F
0..*
contains
1
Gradebook
-gradebookId: String
-attendanceWeight: Float
-midtermWeight: Float
-finalWeight: Float
-isLocked: Bool
+validateWeights(): Bool
+lock(): Void
+unlock(): Void
Semester
- semesterId: String
- semesterName: String
- startDate: Date
- endDate: Date
- registrationStart: Date
- registrationEnd: Date
- cancellationDealine: Date
+ isRegistrationOpen(): Bool
+ isCancellationAllowed(): Bool
has
0..*
1
SemesterGPA
-gpaId: String
-gpa: Float
-totalCredits: Int
+calculate(): Float
0..*